The Trauma-Informed Approach as Glue for HCI and Health Experts

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February 2024

1 Introduction

Research involving those in the fields of both HCI and health is often incredibly valuable; HCI works to improve use and development of technology, thus involving health experts, especially when considering the health of participants, can only lead to positive outcomes. That said, interdisciplinary research involving human computer interaction researchers and those in health can be difficult; these two fields differ in such basic ways that it can be hard to understand the other person's priorities, terminologies, and overarching goals for the project at hand. Scaling these challenges, our team has utilized the Trauma-Informed Approach (TIA) as a bridge for connection and understanding between researchers. Within our project, researchers from the fields of HCI, forensic nursing, psychology, social work, and computer development have come together to accurately apply TIA to an app that plans to collect sensitive data from participants, thus maintaining the mental health of our participants. This overarching goal of mitigating harm to participants through use of TIA has led to intensive collaboration and compromise, and a trauma-informed app that aims to mitigate harm to participants. We posit the Trauma-Informed Approach as a necessary consideration for those conducting research involving sensitive data, and as a point of congruence and team-building for those that conduct interdisciplinary research involving health and HCI. The Trauma-Informed Approach (TIA) was developed for use in a clinical setting, and originally aimed to improve relationships between those who have experiences of trauma and organizations who help them (cite SAMHSA). Most relevant, TIA features six principles (e.g., empowerment, voice, and choice) that assist in protecting the mental health of those who may have experienced trauma. TIA has since been applied to a research setting (cite HCI stuff), though trauma-informed apps and studies have not yet been deployed with participants who may have traumatic experiences.

Our interdisciplinary team has come together by utilizing individual strengths and understanding perspectives of other fields to appropriately apply TIA. Our project will collect longitudinal data through data donation surrounding participant's online to offline sexual experiences using dating apps. The process of utilizing TIA helped us to mitigate participant harm and ensure the continued

mental health of our participants, while still prompting users for meaningful data. By involving social workers and a sexual assault nurse examiner (who is also a forensic nurse), we guaranteed that we were applying TIA accurately, going beyond surface-level applications of the model. Importantly, the primary investigator on this project is a human computer interaction researcher, who led the team, worked to understand TIA, and brought applicable TIA research from the field of HCI to the attention of those involved in development of the project. TIA has been applied to other projects in human computer interaction (e.g., CITE), yet while these papers have suggested and designed for use of TIA, no one yet has applied a TIA data donation application to a population who may have actual experiences of victimization/perpetration (i.e., to the population who may have actually experienced trauma). This new venture necessitated as accurate an application of TIA as possible, as users would be donating sensitive information and reflecting on potentially traumatic events; thus, data donation could incur harm to users through retraumatization. Additionally, two psychology researchers worked to guarantee that the data we receive from participants will be meaningful; in tandem with our HCI researcher, they guaranteed that measurement of these difficult variables, such as consent and sexual violence, were handled carefully, while ensuring that our participants would provide relevant information. The entire design, backed by TIA, has been turned into a usable app by computer developers, and has been subjected to multiple rounds of critique by all faculty involved. The collaboration that has taken place across these five disciplines (i.e., HCI, psychology, forensic nursing, social work, computer development), and across three institutions (i.e., University of Michigan-Flint, Oakland University, and Wayne State), has taught us the importance and benefits of using a multidisciplinary, trauma-informed approach when working with a population with a high rate of trauma. Utilizing TIA as a backdrop for any HCI and health collaboration may both maintain participant's well-being and promote a sense of collaboration and cohesiveness within interdisciplinary teams. Of course, it is crucial to consider the mental health of participants – especially when collecting sensitive data. Projects such as ours pose a risk if they fail to consider the negative impact that they as researchers could inadvertently have on their participants' mental health. Thus, TIA should be considered simply because it may help participants. That said, as it did for us, TIA may also serve as a point of agreement and understanding among those with differing academic backgrounds. Overall, these experiences have prepared us to provide suggestions for others who plan to deploy interdisciplinary projects involving the fields of both HCI and health through implementation of TIA, which may help maximize the strengths of individuals who may enter projects with very different academic backgrounds.